Epigenetic Profiling of Neuroactive Curcumin on Histone Signatures in Breast Cancer using gen3DNet

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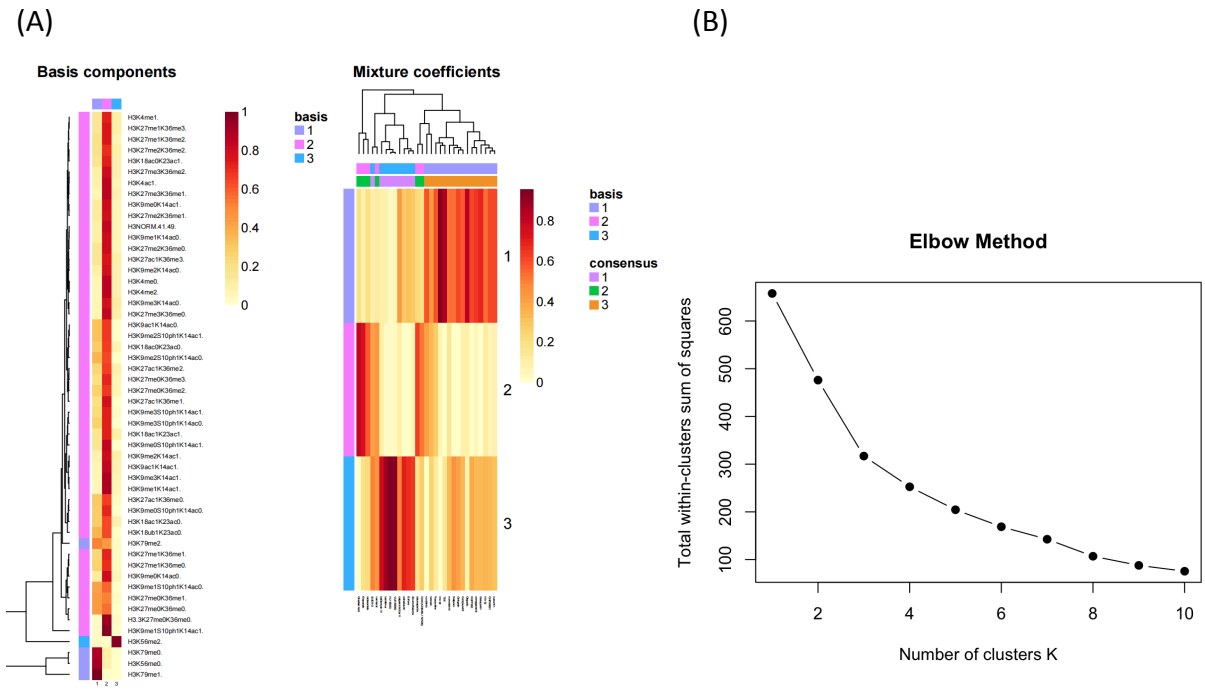

Figure S1. NMF clustering performance of gen3DNet. (A) Depiction of NMF decomposition into basis and loading (mixture) components. Here basis components represent the three functional modules of histone signatures contributed by each neuroactive drug based on their mixture/loading coefficients (higher the coefficient greater the contribution). (B) Determination of optimal cluster (K) using the Elbow (Ward) method. Clustering for different K values (ranging from 2 - 10). X-axis showing number of clusters (K); Y-axis showing total within-cluster sum of squares. The optimal K is at elbow point (where reduction slows down, K=3).

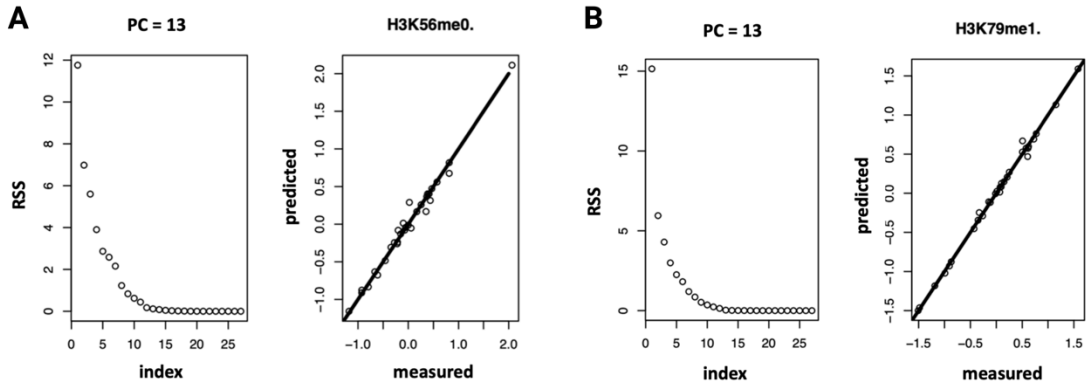

Figure S2. PLSR model performance of gen3DNet. The prediction quality of the PLSR model is shown with the optimal number of principal components (PC) for the two histone codes of interest: (A) H3K56me0 (PC=13) and (B) H3K79me1 (PC=13). The optimal number of components for prediction is identified using residual square sum (RSS) value < 0.05. Overall

effectiveness of the model prediction is depicted showing predicted model (PLSR) on Y-axis and the measured (ground truth) on X-axis, 45 degree line signifies perfect (100%) model prediction.